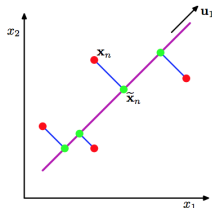
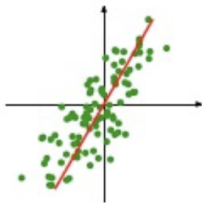


Principal Component Analysis (PCA): Representation

- **Input:** Data $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, $\mathbf{x}_n \in \mathbb{R}^D$, **centered inputs**
- **Output:** Projected data $\{\mathbf{y}_1, \dots, \mathbf{y}_N\}$, $\mathbf{y}_n \in \mathbb{R}^M$, $D \gg M$
- **Projection into subspace:** $\mathbf{U} \in \mathbb{R}^{D \times M}$

$$\mathbf{y}_n = \mathbf{U}^T \mathbf{x}_n, \quad \mathbf{U}^T \mathbf{U} = \mathbf{I}$$

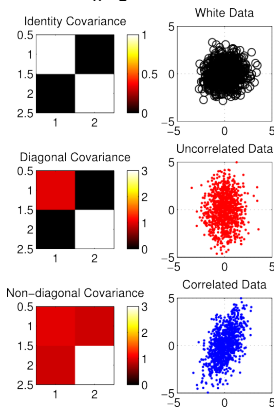
- **Evaluation metric:** many possible metrics yielding the same solution
 - **Derivation 1:** Maximize captured variance
 - **Derivation 2:** Minimize projection error



Covariance Matrix

For 2-dimensional samples $\mathbf{x}_n = [x_{n1}, x_{n2}]^T$, we assume means μ_1 and μ_2 for dimensions 1 and 2.

$$\Sigma = \begin{bmatrix} \sum_{n=1}^N (x_{n1} - \mu_1)^2 & \sum_{n=1}^N (x_{n1} - \mu_1)(x_{n2} - \mu_2) \\ \sum_{n=1}^N (x_{n1} - \mu_1)(x_{n2} - \mu_2) & \sum_{n=1}^N (x_{n2} - \mu_2)^2 \end{bmatrix}$$



Matrix Diagonalization

- Converting a square matrix into a special type of matrix, i.e. diagonal, which shares the same fundamental properties of the underlying matrix
- **Eigen-decomposition theorem:** A square matrix $\mathbf{A} \in \mathbb{R}^{D \times D}$ can be decomposed into

$$\mathbf{A} = \mathbf{P} \mathbf{\Lambda} \mathbf{P}^{-1}$$

- $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_D)$: diagonal constructed from eigenvalues of $\mathbf{A}$
- $\mathbf{P} = \begin{bmatrix} | & | & | \\ \mathbf{e}_1 & \dots & \mathbf{e}_D \\ | & | & | \end{bmatrix} \in \mathbb{R}^{M \times M}$: matrix decomposed from the eigenvectors of $\mathbf{A}$
- The most important advantage of this transformation is that it decorrelates the matrices (i.e., “canonicalizes” the system), therefore some computations might be easier or less expensive

Principal Component Analysis (PCA): Optimization

- Compute covariance matrix

$$\mathbf{S} = \frac{1}{N} \mathbf{X}^T \mathbf{X}, \quad \mathbf{X} = \begin{bmatrix} -\tilde{\mathbf{x}}_1^T - \\ \vdots \\ -\tilde{\mathbf{x}}_N^T - \end{bmatrix}$$

$$\mathbf{U} \in \mathbb{R}^{D \times M}$$

- Diagonalize $\mathbf{S}$, i.e. compute eigenvalues and eigenvectors

$$\mathbf{S} = \mathbf{P} \mathbf{\Lambda} \mathbf{P}^{-1}, \quad \mathbf{P} = \begin{bmatrix} | & & | \\ \mathbf{u}_1 & \dots & \mathbf{u}_D \\ | & & | \end{bmatrix} \in \mathbb{R}^{D \times D}$$

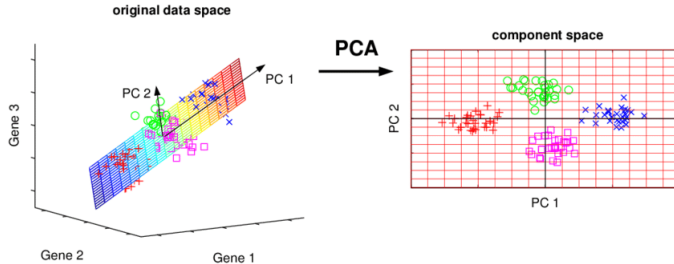
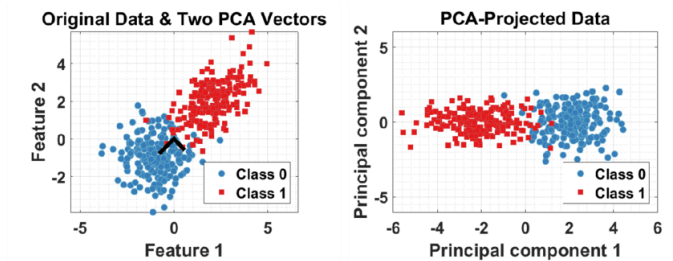
- Use the eigenvectors corresponding to the M largest eigenvalues

$$\mathbf{U} = \begin{bmatrix} | & & | \\ \mathbf{u}_1 & \dots & \mathbf{u}_M \\ | & & | \end{bmatrix} \in \mathbb{R}^{D \times M}$$

Principal Component Analysis (PCA): Algorithm

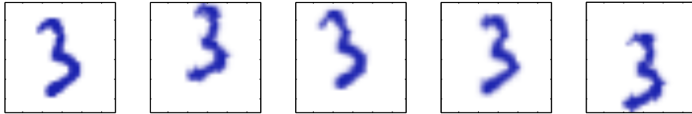
- **Step 0:** Mean normalize input features
- **Step 1:** Compute covariance matrix $\mathbf{S} = \frac{1}{N} \mathbf{X}^T \mathbf{X} = \frac{1}{N} \sum_n \mathbf{x}_n \mathbf{x}_n^T$
- **Step 2:** Diagonalize $\mathbf{S}$ and find eigenvector matrix $\mathbf{P}$
- **Step 3:** Take the first $M \ll D$ eigenvectors or *principal components* (corresponding to the M largest eigenvalues) and form reduced matrix $\mathbf{U}$
- **Step 3:** Project data into reduced space: $\mathbf{z}_n = \mathbf{U}^T \mathbf{x}_n$

Principal Component Analysis (PCA): Example



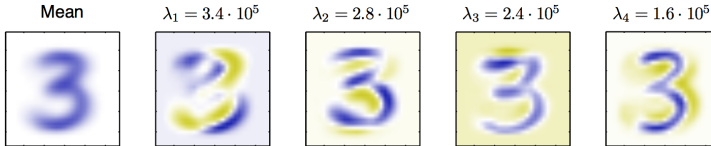
Principal Component Analysis (PCA)

Original Images



Eigenvectors

they look like blurred original images

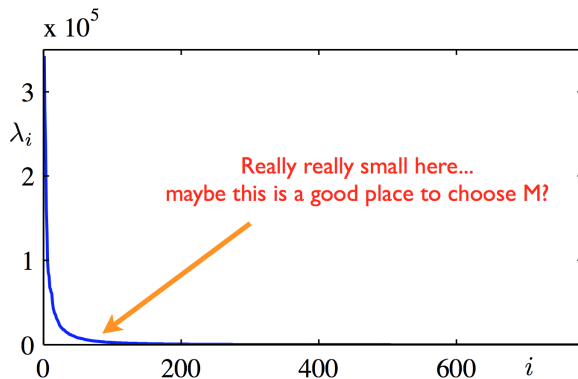


Used to centralize inputs

Principal Component Analysis (PCA)

How to determine the number of principal components M ?

Plot eigenspectrum



$$\frac{\sum_{d=1}^M \lambda_d}{\sum_{d=1}^D \lambda_d} \geq \text{threshold, where common choices are 95\%, 99\%}$$